

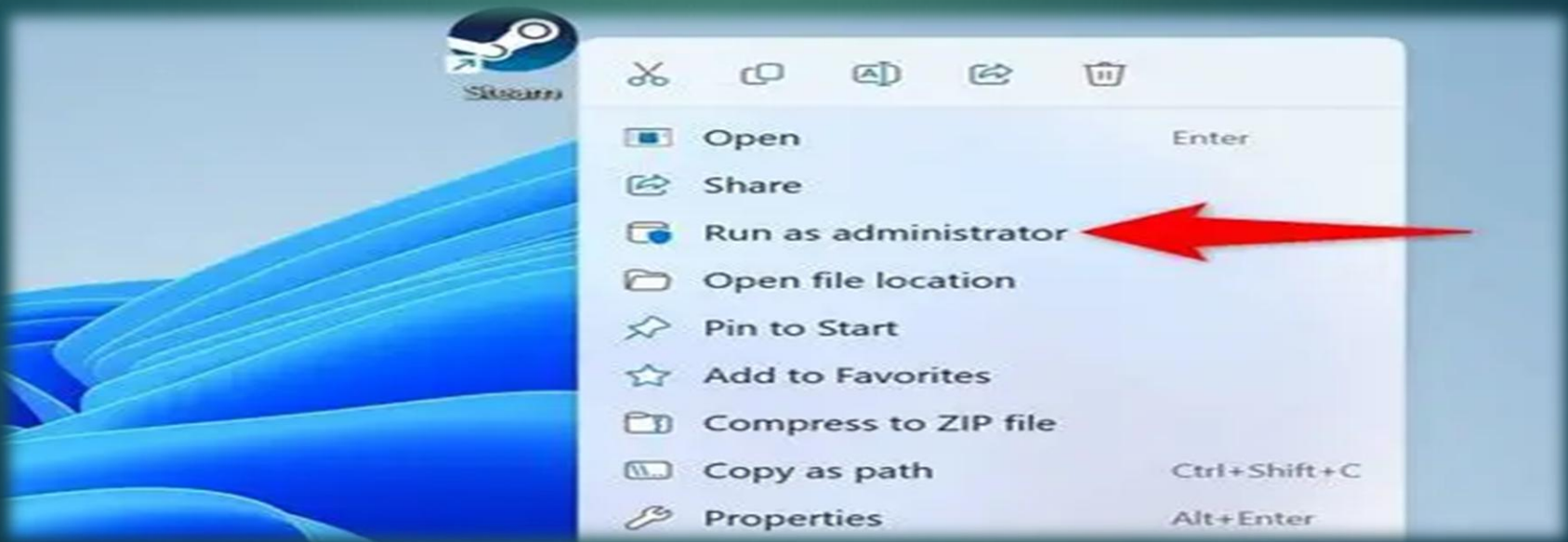


Module 7.3: Windows Configuration and Monitoring

- ENDPOINT SECURITY | CISCO NETWORKING ACADEMY
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Run as Administrator

- ▶ Provides elevated privileges to execute tasks
- ▶ Used for system changes, software installation, and configuration
- ▶ Required when modifying system files or registry settings
- ▶ Works with User Account Control (UAC) for security



Local Users and Domains

- ▶ Local users exist only on a single computer
- ▶ Domain users are managed centrally using Active Directory
- ▶ Domains allow centralized authentication
- ▶ Improves security and access control in organizations

CLI and PowerShell

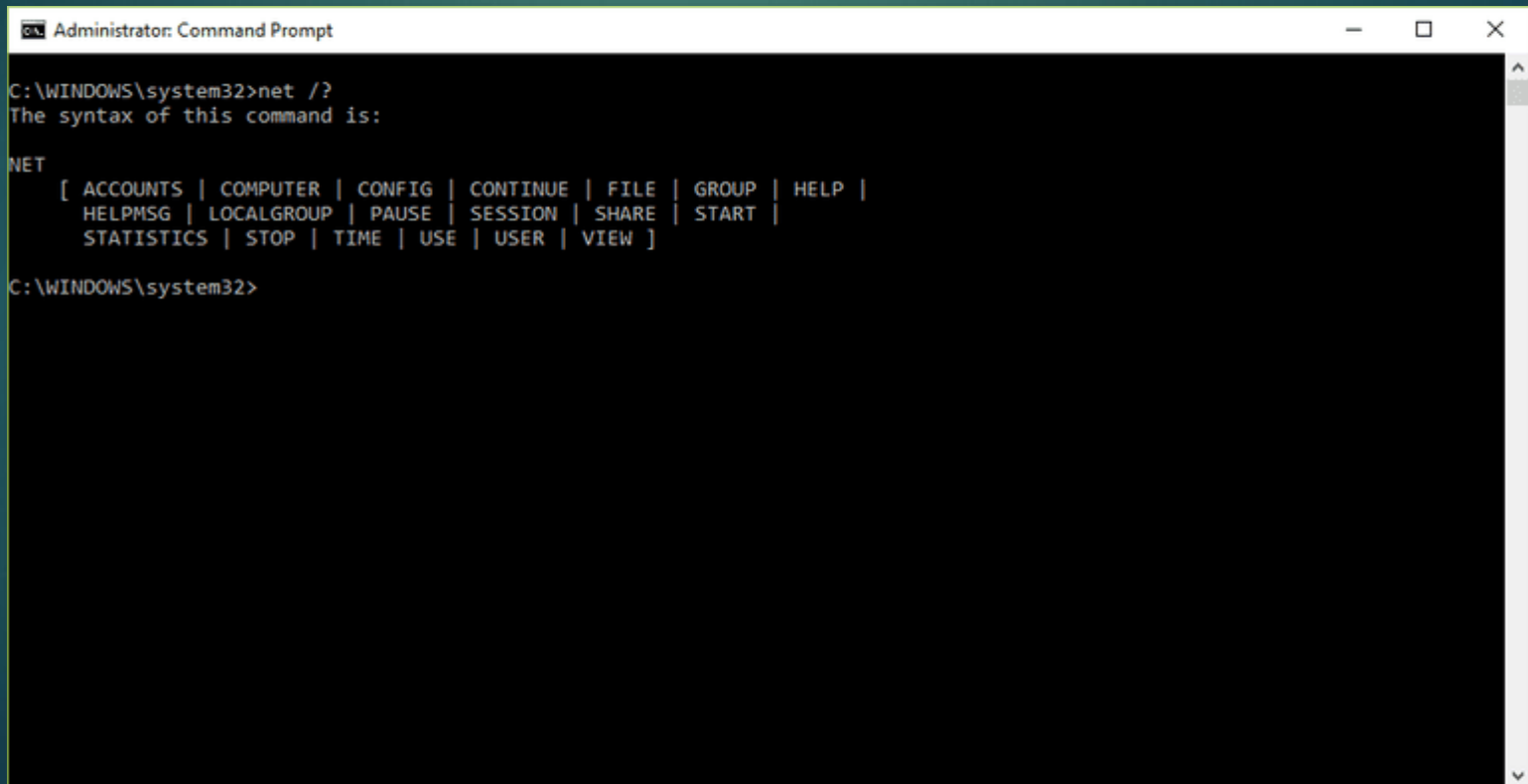
- ▶ CLI allows text-based system management
- ▶ Command Prompt (cmd) handles basic tasks
- ▶ PowerShell supports advanced scripting and automation
- ▶ Used for administration and remote management

Windows Management Instrumentation (WMI)

- ▶ Framework for managing Windows systems
- ▶ Provides system information and monitoring
- ▶ Used for hardware and software management
- ▶ Supports automation and remote administration

The net Command

- ▶ Command-line tool for network administration
- ▶ Examples: net user, net localgroup, net share
- ▶ Manages users, groups, and shared resources
- ▶ Useful for troubleshooting and configuration



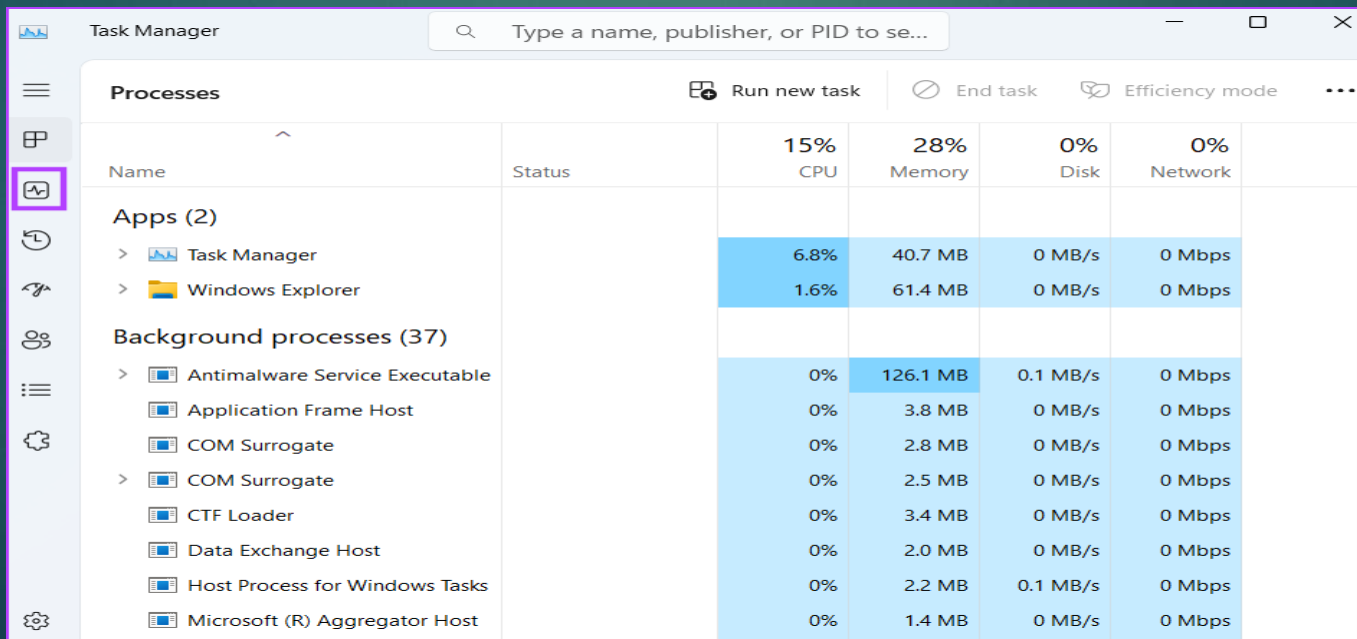
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Administrator: Command Prompt
C:\WINDOWS\system32>net /?
The syntax of this command is:

NET
 [ ACCOUNTS | COMPUTER | CONFIG | CONTINUE | FILE | GROUP | HELP |
  HELPMMSG | LOCALGROUP | PAUSE | SESSION | SHARE | START |
  STATISTICS | STOP | TIME | USE | USER | VIEW ]

C:\WINDOWS\system32>
```

Task Manager and Resource Monitor

- ▶ Task Manager displays running processes
- ▶ Shows CPU, memory, disk, and network usage
- ▶ Resource Monitor gives detailed performance data
- ▶ Helps diagnose performance problems



The screenshot shows the Windows Task Manager application. The title bar reads "Task Manager" with a search bar containing "Type a name, publisher, or PID to se...". The main window is titled "Processes" and includes buttons for "Run new task", "End task", and "Efficiency mode". A sidebar on the left contains icons for Task Manager (highlighted with a red box), Performance, History, Startup, Power, and Settings. The main area displays a table of running processes, categorized into "Apps (2)" and "Background processes (37)". The table columns are Name, Status, CPU, Memory, Disk, and Network. The "Apps (2)" section lists "Task Manager" and "Windows Explorer". The "Background processes (37)" section lists several system processes, including "Antimalware Service Executable", "Application Frame Host", "COM Surrogate", "CTF Loader", "Data Exchange Host", "Host Process for Windows Tasks", and "Microsoft (R) Aggregator Host".

Name	Status	15% CPU	28% Memory	0% Disk	0% Network
Apps (2)					
> Task Manager		6.8%	40.7 MB	0 MB/s	0 Mbps
> Windows Explorer		1.6%	61.4 MB	0 MB/s	0 Mbps
Background processes (37)					
> Antimalware Service Executable		0%	126.1 MB	0.1 MB/s	0 Mbps
Application Frame Host		0%	3.8 MB	0 MB/s	0 Mbps
COM Surrogate		0%	2.8 MB	0 MB/s	0 Mbps
> COM Surrogate		0%	2.5 MB	0 MB/s	0 Mbps
CTF Loader		0%	3.4 MB	0 MB/s	0 Mbps
Data Exchange Host		0%	2.0 MB	0 MB/s	0 Mbps
Host Process for Windows Tasks		0%	2.2 MB	0.1 MB/s	0 Mbps
Microsoft (R) Aggregator Host		0%	1.4 MB	0 MB/s	0 Mbps

Networking

- ▶ Supports wired and wireless connections
- ▶ Uses TCP/IP protocol suite
- ▶ Configuration via Settings or command line tools
- ▶ Common tools: ipconfig, ping, tracert

Accessing Network Resources

- ▶ Access shared folders and printers
- ▶ Use UNC paths (e.g., \\Server\Share)
- ▶ Permissions control user access levels
- ▶ Network drives can be mapped for convenience

Windows Server

- ▶ Designed for enterprise environments
- ▶ Provides services like Active Directory and DNS
- ▶ Manages users, policies, and network resources
- ▶ Supports virtualization and remote access

Conclusion



- ▶ Configuration ensures correct system setup
- ▶ Monitoring tools maintain performance and stability
- ▶ Administrative tools improve control and security
- ▶ Essential knowledge for IT professionals